PolySS: A Compact Filter-and-Refine System for Efficient Approximate Polygon Similarity Search

I. End-to-end Execution Time Analysis

Fig. 1 illustrates the total execution times across datasets, with a grid dimension of 25×25 when PolySS is used. Fig. 1(a) shows the time required to create a global MBR construction phase. This is the first and the shortest phase among all phases. The execution time taken by the other phases is shown in Fig. 1(b). The process begins with the sketching phase, followed by the Hash Construction Phase described. This hash construction phase for input data requires a significant amount of time, the maximum in the whole process. Then comes the phase for the query polygons, such as the query sketching phase, query hashing, and finally, the query refinement phase. The refinement phase consumes the most time for large and complex polygons.

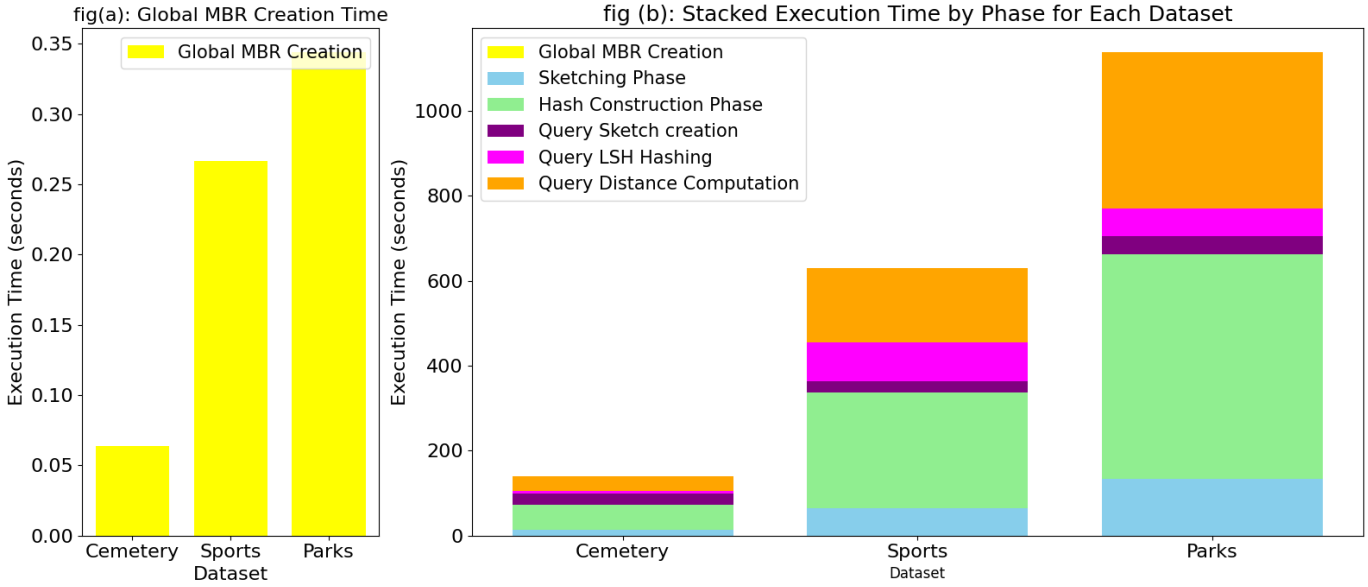


Fig. 1. Execution time breakdown of PolySS for various datasets across 3,000 queries. Figure (a) shows the global MBR construction phase for the entire input data, while Figure (b) depicts the sketching and hash construction phases for the input data, followed by query sketch creation, query LSH hashing, and finally refinement (query distance computation with its candidates) for 3,000 queries.